Project overview: India regional intelligence map

Working name: kerala-live-map (placeholder). Inspired by KW Live (kw-live.com), but built for Indian states and adding forecasting and machine learning. Cost: Rs 0.

1. What the project does

One map per state that brings free public data together: rain, rivers, dams, disaster alerts, air quality, flights, transit, buildings, population, prices and satellite images. Click any ward, taluk or address to see what is happening now, what the models predict next, why, and where each fact came from.

KW Live only shows what is happening now. This project also predicts what happens next and publishes how accurate those predictions were.

2. States and build order

Order	State	Why	Main ML focus
1	Karnataka	Most open data in India	Bengaluru ward flood risk, reservoir forecast, drought, tomato price spikes
2	Kerala	Floods, dams, prices, 2018 flood labels	Dam level forecast, flood and landslide risk, rain-to-price effect
3	Delhi	Live bus GPS, dense air-quality stations	Bus delay, bus bunching, AQI forecast, heatwave risk

3. Features

Map and live layers

1. Live state map: rain, rivers, dams, alerts, air quality, flights, buses, hospitals and shelters, mandi prices
2. Radius filter: show only what is near a point
3. Time slider: replay the past, for example the 2018 Kerala flood or a Bengaluru cloudburst
4. 3D terrain view and globe view
5. Wall mode for a control-room screen
6. Feed health badges: each source shows whether it is live, stale or down
7. Event log and scrolling ticker
8. Daily satellite image layer (NASA GIBS)
9. Satellites passing overhead (CelesTrak)
10. Bus, metro and railway stations; scheduled bus and metro positions from timetables; live buses in Delhi
11. Fire hotspots (NASA FIRMS) and disaster alerts (NDMA Sachet), including lightning
12. Event pins pulled from news by NLP, labelled "preliminary"
13. Flood-exposed roads (OSM roads combined with flood risk)
14. Real-estate projects (RERA) and building outlines (Google Open Buildings)
15. Mandi price ticker
16. District crime totals, if a source becomes available

Place Lens

17. Click a place to see now, risk scores, forecast, history, nearby items and sources
18. Every item labelled official, model prediction or inferred, with a source link and update time
19. Nearest hospital, shelter, police station and fire station
20. Shareable place report (link or PDF) for journalists

Machine learning layers

21. Forecasts: dam and reservoir level, ward flood risk, landslide risk, drought, AQI, bus delay, price spikes
22. "What if" simulator: enter a rain amount and the flood-risk map redraws
23. Explanation for each prediction (SHAP)
24. Uncertainty ranges on forecasts (40-member weather ensemble)
25. Model accuracy page plus a daily backtest of forecasts against what happened
26. Anomaly feed: unusual rain, dam jumps and price jumps found automatically

People, risk and climate

27. People at risk inside flood and landslide zones (WorldPop)
28. Safe route to a shelter that avoids flood-risk roads
29. Heatwave and heat-index layer
30. Crop health from satellite vegetation index
31. Rainfall trend since 1901 for each district
32. Climate outlook to 2050 (CMIP6)
33. Monsoon onset and progress tracker
34. Past-disaster replay

Ask, alerts and memory

35. Ask: questions by text or voice, answered from the data with citations
36. Kannada, Malayalam, Hindi and English
37. Watch a place: alerts when risk is high, a dam crosses a level or a price spikes
38. Telegram alert bot and web push
39. Daily brief: an automatic morning summary for each state
40. City Memory: every snapshot is stored, so you can see what was known and forecast on any past date

Data and access

41. Analysis dashboards: rain, dams, prices, bus coverage
42. Data quality dashboard
43. Open data download and public API
44. Installable mobile app (PWA) that works offline
45. Embeddable widgets for news sites
46. State comparison view

4. Use cases

User	Question it answers
Residents	Is my area at flood or landslide risk this week?
Farmers and traders	Will tomato or onion prices jump after this rain?
Disaster teams and NGOs	Which wards or taluks need help first, and how many people live there?
Journalists	What happened where, with the official source one click away?
Researchers and students	Where can I get clean history data and tested models?
Delhi commuters	Is my bus late, and how bad is the air on my route?
City planners	Which wards have poor bus coverage, and which roads flood?
Recruiters	Can this person do data engineering, analysis, ML and LLM work in one real project?

5. How it covers each discipline

Discipline	What the project does
Data engineering	Daily pipelines from about 30 sources, cleaning, quality checks, versioned storage, snapshots
Data analysis	Trends, comparisons, dashboards, the bus coverage study, the 2018 flood study
Data science	Correlation and causal tests (rain to price), spatial statistics, anomaly detection, uncertainty
Machine learning	Forecasting, classification, spatial interpolation, explainability, transfer between states
Deep learning	LSTM or Temporal Fusion Transformer forecasts, U-Net satellite flood mapping, NLP on Indian-language news
LLM	Retrieval-augmented Ask with citations, plus evaluation of answer quality

6. Not possible in India (even with money)

Live road closures, live police and fire calls, public traffic cameras, police radio, live KSRTC buses, and groundwater levels (the CGWB site failed in testing).

7. Rs 0 tech stack

Need	Free option
Pipeline	GitHub Actions cron (public repository)
Storage	Parquet on Cloudflare R2 (10 GB free) or Hugging Face Datasets
Queries	DuckDB, also DuckDB-WASM in the browser
Website	Cloudflare Pages
Map	MapLibre GL, OpenFreeMap, AWS terrain tiles, NASA GIBS
ML training	Own laptop, Kaggle (free GPU), Google Colab
Model serving	Predictions computed daily and published as static files (no server)
Ask	Free LLM tier (Gemini, Groq or Cloudflare Workers AI) plus local embeddings
Voice	Browser Web Speech API
Alerts	Telegram Bot API, web push